

PADMESH MK

padmeshmk@gmail.com | 8825640679 | Chennai, Tamil Nadu

PROFESSIONAL SUMMARY

Fourth-year B.E. Computer Science student specializing in Cybersecurity with hands-on experience in Python, embedded systems, and secure application development. Proficient in penetration testing tools including Kali Linux, Burp Suite, and Metasploitable, with active practice on Hack The Box. Seeking an internship to contribute technical skills in a professional environment.

TECHNICAL SKILLS

Programming Languages: Python

Frameworks & Tools: Flask, PyTorch, FastAPI, Arduino IDE

Cybersecurity Concepts: Multi-Factor Authentication (MFA), TOTP, Network Security, Penetration Testing, Vulnerability Assessment

Cybersecurity Tools: Kali Linux, Metasploitable, Burp Suite, Hack The Box (HTB)

PROJECTS

Vision Transformer (ViT) Based Object Detection with Real-Time Alerts

Tools: Python, PyTorch, HuggingFace Transformers (YOLOS), OpenCV, FastAPI, WebSocket, Uvicorn | 2025 (In Progress)

- Built a real-time object detection system using YOLOS (Vision Transformer-based architecture) via HuggingFace Transformers and PyTorch, overcoming CNN limitations with occlusions and cluttered backgrounds through global self-attention.
- Integrated the model into a FastAPI backend with async MJPEG streaming and WebSocket alert broadcasting for low-latency inference on live video streams, with CUDA GPU support.
- Implemented configurable target detection with debounced real-time alerts and automatic snapshot saving, using thread-safe asyncio queues to decouple camera inference from the event loop.

Multi-Factor Authentication (MFA) for Cloud Applications

Tools: Python, PyOTP, Flask, TOTP, Google Authenticator | 2025

- Designed and implemented a two-factor authentication system for cloud applications using Time-Based One-Time Passwords (TOTP) with PyOTP and Flask.
- Integrated QR-code-based pairing with Google Authenticator to enable seamless user enrollment and secure login.
- Reduced risk of unauthorized access by enforcing password + TOTP verification, demonstrating practical application of identity and access management (IAM) principles.

Smoke and Gas Detector using MQ-5 Sensor and Arduino

Tools: Arduino UNO, MQ-5 Gas Sensor, Buzzer, LED, Python | 2023

- Built a real-time gas leak and smoke detection system using the MQ-5 sensor interfaced with Arduino UNO.
- Programmed alert logic in Arduino IDE to trigger buzzer and LED indicators upon detection of hazardous gas levels.
- Demonstrated core skills in embedded systems design, sensor calibration, and real-time monitoring.

EDUCATION

B.E. Computer Science and Engineering With Specialization in Cybersecurity

Sathyabama Institute of Science and Technology, Chennai | Expected 2026 | CGPA: 6.68/10

Higher Secondary Certificate (HSC)

Aristo Public School, Thirupathiripuliyur, Cuddalore | 2022 | 68.6%

Secondary School Leaving Certificate (SSLC)

Aristo Public School, Thirupathiripuliyur, Cuddalore | 2020 | 62%